Inference-Time Model Steering via Predictive-State Intervention: A Survey

Renhao Xue **Rui Wang** **Yawei Wang** **Yueying Cui** **Yiyue Qian**
Praneetha Vaddamanu **Huan Song** **Hannah R Marlowe**
AWS Generative AI Innovation Center
{lukasxue, rwngamz, yawenwan, yueyinc, iamyiyue,
lpranvad, huanso, marloweh}@amazon.com

Abstract

Inference-time steering controls large language models by intervening on their internal computations during generation, with strength tunable and the intervention reversible per request. Recent work spans activation steering, sparse-feature interventions, classifier-guided decoding, and external-memory distribution interpolation, but these threads have developed independently. We unify them as *perturbations to the predictive state of an unmodified base LM*, organized by *intervention location* (latent, logit, distribution; decoding as boundary cases) as the primary dimension that determines expressiveness, side-effect surface, and required model access. We use the *perturbation ratio* ρ to organize stability across methods within each location. We synthesize findings from recent benchmarks (AxBench, SAEBench), frame prompting as the practical baseline rather than as part of the steering family, and chart open problems in perturbation geometry, composability, and safety governance.

1 Introduction

Large language models (LLMs) have achieved remarkable capabilities across diverse tasks, from open-ended generation to complex reasoning (Brown et al., 2020; Chowdhery et al., 2023; Touvron et al., 2023). Yet controlling their behavior remains a central challenge. Conventional control relies on *fine-tuning* or *reinforcement learning from human feedback* (RLHF) to shape model outputs (Ouyang et al., 2022; Bai et al., 2022). While effective, these approaches commit behavioral changes at training time: adjusting a constraint typically requires retraining or selecting from a set of learned parameter blocks, rather than applying a lightweight external signal per request.

An alternative paradigm has emerged: *inference-time steering*, where a model is guided during generation through interventions (activation vectors, logit biases, retrieval signals) that can be dynamically adjusted without retraining. The defining criteria that separate steering from parameter-efficient fine-tuning (PEFT) methods like LoRA (Hu et al., 2022) and prefix tuning (Li and Liang, 2021) are: *(1) tunability at serving time via an external per-request signal applied to the fixed base computation, rather than by selecting or training learned parameters; and (2) reversibility, where interventions are applied per-request and removed without state changes to the base model* (see Table 3 in Appendix B for a detailed comparison). Interpretability is a frequent but non-defining attribute.

The literature has grown rapidly (App. C), spanning activation steering (Turner et al., 2024; Rimsky et al., 2024; Li et al., 2023a), sparse-feature interventions (Chalnev et al., 2024; Cho et al., 2026), classifier-guided decoding (Krause et al., 2021; Yang and Klein, 2021; Liu et al., 2021), and external-memory distribution interpolation (Khandelwal et al., 2020; Grave et al., 2017). These threads have developed largely in isolation, using different vocabularies that obscure their shared structure and make systematic comparison difficult.

This survey makes three main contributions:

1. **Unified framework organized by intervention location.** We formalize inference-time steering as an intervention operator on a predictive state, organized by *intervention location* as the primary dimension, with location-conditioned attributes (signal source, the perturbation ratio ρ for stability, and interpretability) (§3).
2. **Method landscape with case studies and within-family comparisons.** We survey approximately 50 methods across the four locations (latent, logit, distribution; decoding as boundary), with detailed case studies of representative methods at each location and within-family comparison tables for SAE-based, stability-oriented, and logit-level variants that surface the design choices each family faces (§4, App. I).

3. **Practical guidance and research roadmap.** We synthesize benchmark lessons, extract actionable principles for practitioners, identify common failure modes, and chart what the field needs to accelerate progress (§5).

2 Background

2.1 Transformer-Based Language Models

A transformer-based LM (Vaswani et al., 2017) with parameters θ generates text autoregressively. At step t, the input context x and prior tokens $y_{<t}$ pass through L layers, each updating a hidden state $h_{\ell,t} \in \mathbb{R}^d$. The final hidden state is projected to logits $\ell_t = W_{\text{out}}\, h_{L,t} + b$, yielding the next-token distribution $p_\theta(y_t \mid y_{<t}, x) = \text{softmax}(\ell_t/\tau)$, after which a decoding algorithm $\mathcal{D}$ selects the next token. The sequence $h_{0,t}, \dots, h_{L,t}$ is called the *residual stream*: because of residual connections, each layer adds an update to this stream, so adding a vector at layer ℓ propagates through all subsequent layers without changing weights. Notation is summarized in Appendix E.

2.2 Key Concepts

Several concepts from interpretability research underpin the steering methods in this survey:

Linear representation hypothesis. Evidence suggests that many semantic concepts (e.g., "truthfulness," "toxicity," "sentiment") correspond to approximately linear directions in a model's activation space (Burns et al., 2023; Zou et al., 2023; Taimeskhanov et al., 2026; Park et al., 2023; Marks and Tegmark, 2023). If a concept is encoded as a direction v in $\mathbb{R}^d$, then adding αv to the hidden state can amplify or suppress that concept during generation. The hypothesis is contested: Engels et al. (2025) show that some concepts (e.g., days of the week, months of the year) are encoded as multidimensional manifolds (a circle, in those cases) that no single direction can capture, limiting contrastive-vector methods on such concepts.

Mechanistic interpretability and sparse autoencoders. Mechanistic interpretability reverse-engineers how transformer circuits implement specific behaviors (Elhage et al., 2021; Wang et al., 2023), with findings tracing some factual associations to MLP-layer mechanisms (Meng et al., 2022; Geva et al., 2021). A central tool for fine-grained intervention is the *sparse autoencoder* (SAE) (Anthropic, 2023): because individual neurons are *polysemantic*, SAEs learn a dictionary of $k \gg d$ approximately monosemantic *features* by training an encoder/decoder pair to decompose each hidden state $h \in \mathbb{R}^d$ into a sparse combination of dictionary vectors (Gao et al., 2025).

2.3 Related Surveys

Adjacent surveys focus on controllable generation (Zhang et al., 2023; Prabhumoye et al., 2020), knowledge editing (Zhang et al., 2024), representation engineering (Bartoszcze et al., 2025), or broader inference-time improvement (Dong et al., 2024). The closest competitor, Zhang et al. (2026), organizes actionable mechanistic interpretability around *Interpretable Objects* (what is targeted) rather than intervention location, and does not unify logit and external-memory interventions with latent steering. Our differentiation is threefold: (i) one operator $z_t' = z_t + \delta$ across latent, logit, and distribution-level interventions; (ii) intervention location as the primary organizing axis, with location-conditioned attributes (ρ, signal source, interpretability) as secondary dimensions; and (iii) coverage of 2024–2026 methods (sparse-feature, geometric, adaptive, ODE-based) that postdate prior surveys. Appendix D gives a row-by-row comparison; Appendix F lists inclusion criteria.

3 A Unified Framework for Inference-Time Steering

We unify these threads along four axes: a taxonomy organized by *intervention location* (§3.1); a common operator $z_t' = z_t + \delta$ over a predictive state (§3.2); location-conditioned attributes covering expressiveness, the stability quantity ρ, and interpretability (§3.3); and composition across multiple interventions (§3.4).

3.1 Intervention Location

The autoregressive pipeline reveals a sequence of increasingly external intervention points (Figure 1), from internal representations to the final token-selection rule. Three locations admit the operator $z_t' = z_t + \delta$ on a predictive state; decoding does not and is treated as a boundary case:

1. **Latent states** $h_{\ell,t}$: modifications to residual-stream activations at any layer ℓ and position t. These propagate through all subsequent layers, enabling broad behavioral changes (§4.1).
2. **Logits** ℓ_t: direct modifications to output scores before the softmax. These affect only the cur-

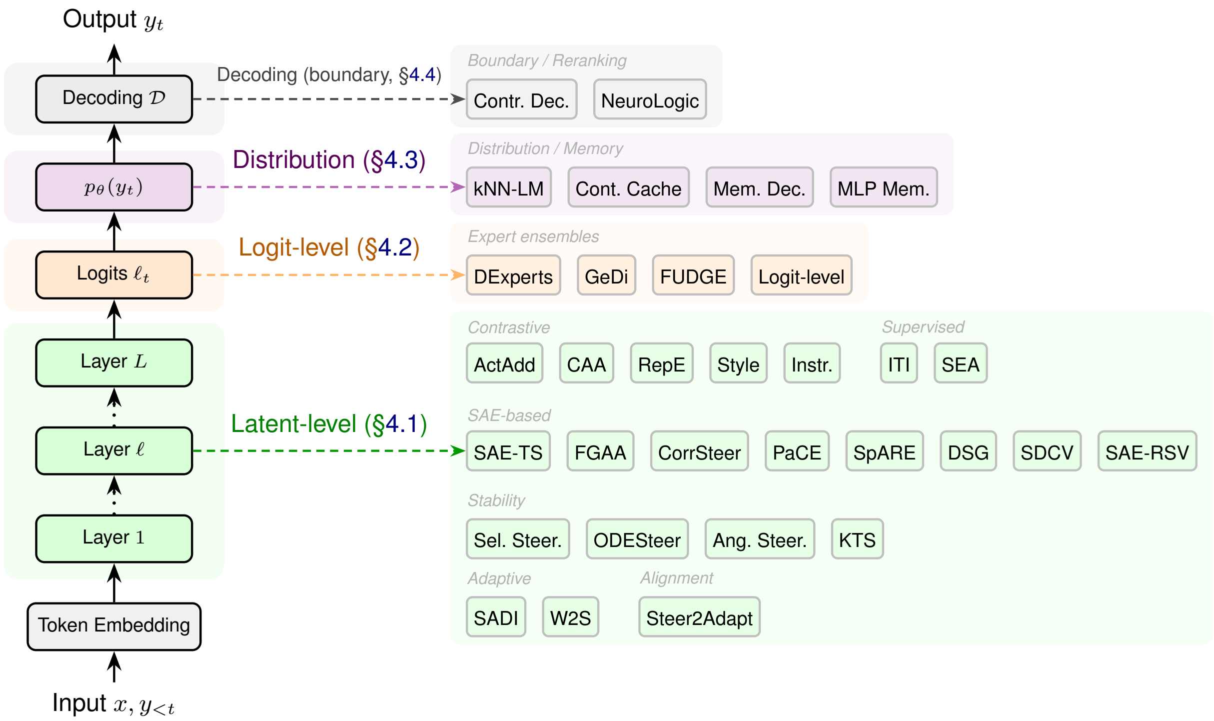

Figure 1: The autoregressive generation pipeline (left) and the method landscape organized by intervention location (right). Dashed arrows connect each pipeline stage to the methods that intervene at that location. Latent methods are grouped into four rows: contrastive/supervised, SAE-based, stability-oriented, and adaptive/alignment. Decoding-level methods are treated as boundary cases (§4.4). Full taxonomy in Table 2 (Appendix A).

rent token and do not propagate through internal computations (§4.2).

3. **Output distributions** $p_\theta(y_t)$: mixing the model's next-token distribution with an external distribution (from a retrieval datastore or parametric memory) after the forward pass, without modifying the model's input (§4.3).
4. *Boundary: decoding objective* $\mathcal{D}$: changes to how tokens are selected or sequences are scored. These modify $\mathcal{D}$ rather than z_t, so the operator and ρ do not apply (§4.4).

Table 2 (Appendix A) maps representative methods at each location to the attributes used throughout the survey: *signal source* (how δ is computed), *stability* (governed by ρ; §3.3), and *interpretability* (how transparent the intervention is to a user); the table also reports granularity (behavioral level of control) for completeness.

3.2 The Intervention Operator

We define a *predictive state* as any intermediate object along the generation pipeline that determines the model's next-token output: a hidden representation $h_{\ell,t} \in \mathbb{R}^d$, a logit vector $\ell_t \in \mathbb{R}^{|V|}$ over the vocabulary V, or a probability distribution p_t over V. Let z_t denote such a predictive state at decoding step t. Steering at the first three locations applies an *intervention operator*:

$$z_t' = z_t + \delta(z_t, x, y_{<t}; \phi), \tag{1}$$

where δ is the steering perturbation, optionally parameterized by ϕ. The generated token is then selected by a decoding algorithm: $y_t \sim \mathcal{D}(z_t')$. Decoding-level methods (§4.4) instead modify $\mathcal{D}$ itself while leaving z_t unchanged.

The additive form gives heterogeneous methods three comparable handles: *magnitude* (read off as the norm ratio $\rho = \|\delta\|/\|z_t\|$; §3.3), *reach* (how far δ propagates: through $L-\ell$ nonlinear blocks for latent edits, through the final softmax only for logit edits, through neither for distribution mixing; §3.3), and *composition* (combining multiple interventions reduces to summing δ's, with ρ accounting; §3.4).[1]

3.3 Location-Conditioned Attributes

Location is primary because it determines the trade-offs every method inherits. The remaining attributes (expressiveness, stability via ρ, and interpretability) are not free parameters: each takes a characteristic range at each location.

[1]Non-additive operations (rotations, ablations, multiplicative reweighting) fit the same form by setting $\delta(z_t, x, y_{<t}) = f(z_t) - z_t$ for an appropriate transformation f, so magnitude, reach, and composition still apply.

Expressiveness vs. side effects. Latent edits at earlier layers propagate through $L - \ell$ nonlinear blocks, yielding context-dependent logit changes that no fixed bias can replicate; conversely, any logit-level bias in the column space of W_{out} admits a final-layer latent counterpart via the pseudoinverse (App. G). This asymmetry produces the practical **expressiveness vs. side-effect trade-off**: latent methods are more expressive but carry larger side-effect surfaces; logit methods are more predictable but less expressive. Empirically, expressiveness does not always translate into better steering, especially on concept-level tasks (Wu et al., 2025b) (§5); broad behavioral shifts favor latent expressiveness, while narrow token-level constraints favor logit predictability.

Stability: the perturbation ratio ρ. Early activation steering tracked relative norms only informally (Turner et al., 2024; Rimsky et al., 2024), motivating geometric methods that bound magnitude by construction: norm-preserving rotation (Vu and Nguyen, 2025; Dang and Ngo, 2026) and multi-step bounded integration (Zhao et al., 2026b). We unify these threads under the *perturbation ratio* ρ. For vector-valued states, $\rho = \|\delta\|_2 / \|z_t\|_2$; for distribution-level methods, the analogous quantity is total variation $\mathrm{TV}(p_t, p'_t) \leq \lambda$; decoding-level methods modify $\mathcal{D}$ rather than z_t, so ρ does not apply. Because these quantities are location-specific, we do not use ρ for cross-location ranking. Within each location, larger perturbations yield stronger control but risk degrading output quality, and methods that bound ρ by construction tend to retain capabilities better (Stickland et al., 2024; Taimeskhanov et al., 2026). ρ is necessary but not sufficient: at matched magnitude, random directions can collapse generation while task-relevant directions preserve it (App. H), so methods that also constrain geometry (rotations) or support (sparse SAE features (Chalnev et al., 2024)) offer the strongest capability retention. Table 1 classifies methods by how they manage ρ.

Interpretability. Interpretability also varies systematically with location: latent SAE features carry candidate semantic labels, dense activation steering exposes a direction without per-dimension meaning, logit biases are vocabulary-inspectable, and retrieved exemplars in distribution-level methods provide explicit provenance. Per-location detail in §4; Table 2 (App. A) summarizes.

Strategy	Methods	Mechanism
Unconstrained	ActAdd, CAA	ρ input-dependent
Geometric	Angular, Sel. Steer., ODESteer	Rotation or small steps
Support	SAE methods	Sparse feature subset
Logit	GeDi, FUDGE, DExperts	Affects output entropy only

Table 1: Strategies for managing ρ. Methods that constrain norm, geometry, or support (rows 2–3) tend to retain capabilities better than unconstrained additive baselines.

3.4 Composition

Practical deployments sometimes combine multiple interventions. Three patterns arise. *Additive composition* sums signals at the same point, $\delta = \sum_{i=1}^{n} \alpha_i v_i$: simple, but per-property efficacy degrades when vectors share components, and the combined ρ grows roughly with $\sqrt{n}$ for orthogonal vectors and faster otherwise (Scalena et al., 2024). *Sequential composition* chains interventions across locations (e.g., latent tone steering plus logit format constraints), avoiding the shared-direction problem at the cost of ordering effects: an upstream latent edit changes the logits a downstream classifier sees, so commuting the order is not generally safe. *Gated composition* learns input-dependent weights $w_i(z_t)$ (Han et al., 2026), which empirically reduces interference at the cost of an extra training stage; related learned-composition approaches generate vectors directly via hypernetworks (Sun et al., 2025) or operate in input-token space (Radevski et al., 2026). The literature reports per-method composition results (Scalena et al., 2024; Han et al., 2026) but few systematic studies across heterogeneous families; understanding when compositions interact destructively remains open (§5).

4 Method Landscape: Steering by Intervention Location

We organize the steering landscape by *intervention location*, the primary dimension introduced in §3. Table 2 (App. A) gives the full method-level details. The latent subsection is the longest because it covers the largest and most contested literature; the logit, distribution, and decoding subsections are correspondingly shorter.

4.1 Latent-Level Methods

Latent-level methods modify hidden states $h_{\ell,t}$ inside the residual stream. They specialize the intervention operator (Eq. 1) by setting $z_t = h_{\ell,t}$, so $h'_{\ell,t} = h_{\ell,t} + \delta$ at the chosen layer ℓ. Because perturbations propagate through all subsequent layers, these methods are the most expressive (§3.3) but also carry the largest side-effect surface and the greatest ρ-management burden.

We organize this subsection by how δ is constructed and constrained: training-free contrastive extraction, supervised probe directions, learned sparse features, and stability-oriented methods that constrain ρ; we close with interpretability.

Training-free contrastive extraction. The simplest approach extracts a steering vector from *contrastive prompt pairs* without any optimization. Given positive prompts $\{x_i^+\}$ and negative prompts $\{x_i^-\}$, the steering vector is the mean activation difference at a chosen layer ℓ:

$$v = \frac{1}{N} \sum_{i=1}^{N} \left[h_\ell(x_i^+) - h_\ell(x_i^-) \right]. \quad (2)$$

The intervention is then $\delta = \alpha v$, applied at layer ℓ during generation.

Contrastive Activation Addition (Rimsky et al., 2024) operationalizes this approach by averaging over hundreds of contrastive example pairs and evaluating on behavioral dimensions such as sycophancy and refusal in Llama 2. The vector can be computed offline in minutes and applied during inference as a single addition per layer, incurring negligible latency overhead. However, CAA provides no mechanism to control the perturbation magnitude relative to the model's activation norms, leading to ρ values that vary widely across inputs and causing unpredictable capability degradation. ActAdd (Turner et al., 2024) and RepE (Zou et al., 2023) share the same contrastive extraction mechanism (Eq. 2) and unconstrained ρ; they differ only in whether v comes from a single pair, N pairs, or PCA over activation differences. Extensions include style vectors (Konen et al., 2024) and instruction-following steering (Stolfo et al., 2025).

Supervised probe directions. Supervised methods use labeled data to find more precise intervention directions than contrastive means.

Inference-Time Intervention (Li et al., 2023a) trains *linear probes* on labeled truthfulness data to identify, for each attention head, the direction in its activation space that best separates truthful from untruthful model states. It then selects the top-K heads by probe accuracy and shifts their activations along the learned direction: $\delta = \alpha\sigma\hat{w}$, where $\hat{w}$ is the unit probe direction and σ is the standard deviation of activations along that direction.

The key insight is that labeled data enables ITI to find directions that are both more precise and lower-norm than contrastive means, achieving stronger steering at smaller ρ than unsupervised contrastive methods. The limitation is that labeled data is required for each target behavior.

Spectral Editing of Activations (SEA; Qiu et al. 2024) identifies a subspace of relevant directions by analyzing the principal axes of variation in activation differences, offering a middle ground between full-vector addition and per-feature editing.

Learned sparse features. SAE-based methods operate in the disentangled feature space of sparse autoencoders (§2), enabling finer-grained and more interpretable control than dense vector methods.

CorrSteer (Cho et al., 2026) selects which SAE features to steer by correlating each feature's activation with a binary task-success signal during generation, then modifies only the top-m correlated features. This introduces *generation-time feature selection* and the *Side Effect Ratio (SER)* metric for quantifying unintended behavioral changes.

The core challenge across all SAE steering methods is *feature selection*: which of the $k \gg d$ features to modify, and by how much. Methods differ in selection criterion: optimization against off-target features (SAE-TS, FGAA), correlation with task signal (CorrSteer), mutual information (SpARE), dynamic classifiers (DSG), projection (PaCE), or denoising (SDCV, SAE-RSV). Table 9 (Appendix I.1) provides a full comparison.

SAEBench (Karvonen et al., 2025) evaluates SAEs across eight metrics and finds that improvements on common proxy metrics (reconstruction loss, sparsity) do not reliably translate to practical steering performance. Comparative studies (Xie, 2025) show that single top-1 SAE latents with token-wise decaying steering can outperform both top-k methods and mean activation difference baselines. SAE-based denoising methods (Zhao et al., 2026a; Wang et al., 2025a) further improve steering success rates by filtering task-irrelevant features from concept vectors.

Stability (ρ Management). Several geometric approaches bound ρ during inference. Selective Steering (Dang and Ngo, 2026), instead of adding a vector δ to the hidden state, *rotates* the hidden state within a subspace. Let $\hat{v}$ be the unit steering direction. The intervention applies a rotation matrix R_θ in the plane spanned by $h_{\ell,t}$ and $\hat{v}$:

$$h'_{\ell,t} = R_\theta\, h_{\ell,t}, \quad \text{where } \|h'_{\ell,t}\| = \|h_{\ell,t}\|. \tag{3}$$

Because rotation preserves the vector's norm by construction, the perturbation ratio is bounded: $\rho \leq 2\sin(\theta/2)$ for rotation angle θ, preventing the activation norm explosion that causes generation collapse. The key result is zero perplexity violations across 9 tested models.

ODESteer (Zhao et al., 2026b), rather than constraining the geometry of a single perturbation, breaks it into K small steps: $z^{(k+1)} = z^{(k)} + \eta\, g(z^{(k)}, c)$, where g includes a barrier term pushing the trajectory away from constraint boundaries. Because each step has small $\|\delta^{(k)}\|$, the trajectory stays close to the activation manifold. ODESteer reports +5.7% on TruthfulQA and +2.4% on RealToxicityPrompts over single-step baselines.

Angular Steering (Vu and Nguyen, 2025) uses rotation (like Selective Steering) but with a calibrated angle θ tuned to the task.

A complementary line trains for stability rather than enforcing it at inference: KL-then-Steer (Stickland et al., 2024) fine-tunes the LM to minimize KL divergence between steered and unsteered outputs on benign inputs, reducing side effects at matched efficacy. Adaptive extensions address the orthogonal limitation that a static vector cannot adapt to varying contexts: SADI (Wang et al., 2025d) selects the steering direction, and Where-to-Steer (Gadgil et al., 2026) the intervention layer, based on input semantics; while Dynamic Activation Composition (Scalena et al., 2024) and Steer2Adapt (Han et al., 2026) enable multi-property control by composing vectors with learned or information-theoretic weighting. Table 10 (App. I.2) summarizes the stability mechanisms.

Interpretability. Latent methods span the interpretability spectrum, and the level at which δ is constructed determines how much of it is human-readable. SAE features admit auto-interpretability pipelines that assign natural-language labels to individual dictionary atoms (Anthropic, 2023), with downstream tools tracing how features evolve and compose across layers (Laptev et al., 2025); reliability is uneven, however, and proxy metrics such as reconstruction loss do not reliably predict label quality or steering performance (Karvonen et al., 2025). Supervised probe directions inherit the label of the trained classifier ("the truthfulness direction" for ITI), giving directional semantics without per-dimension meaning, while training-free contrastive vectors (CAA, ActAdd) carry only the implicit semantics of their contrastive set. Dense activations themselves are polysemantic at the neuron level (Anthropic, 2023), which is why interpretability at this location effectively requires either an SAE dictionary or a labeled probe.

4.2 Logit-Level Methods

Logit-level methods modify output scores ℓ_t directly, specializing the intervention operator (Eq. 1) with $z_t = \ell_t$ so that $\ell'_t = \ell_t + \delta$. Because there are no downstream layers to destabilize, ρ affects only output entropy, not internal representations, making these methods more stable than latent interventions. The trade-off is reduced expressiveness: a fixed logit bias cannot produce the context-dependent, input-adaptive changes that latent methods achieve (§3.3).

DExperts (Liu et al., 2021) trains an *expert* LM on desirable text and an *anti-expert* LM on undesirable text. At each generation step, the base model's logits are adjusted:

$$\ell'_t = \ell_t + \alpha(\ell_t^{\text{expert}} - \ell_t^{\text{anti}}). \tag{4}$$

The contrastive formulation cancels shared linguistic structure and amplifies attribute-specific signals. Because the intervention operates on logits, it largely avoids the internal activation distribution shift problems that plague latent methods. GeDi (Krause et al., 2021) and FUDGE (Yang and Klein, 2021) replace the expert/anti-expert pair with auxiliary classifiers, deriving δ from a Bayes-factor ratio or a future-attribute discriminator respectively; all three methods share the additive logit form and differ only in how the bias is computed (Table 11, App. I.3).

Stability. At large α, the failure mode is softmax saturation: probability mass collapses onto a small set of tokens, producing degenerate but fluent text. The effective bias entering the softmax is δ/τ, so ρ couples with sampling temperature and is not directly comparable across settings that differ in τ. An et al. (2026) obtain consistent multi-task control (writing complexity, formality, toxicity) with z-normalized log-odds biases.

Interpretability. The bias vector is defined over the vocabulary, so each promoted or suppressed token is inspectable in natural language without an SAE dictionary or feature labeler. Logit audits therefore expose *which* tokens an intervention favors, but not *why*: the auxiliary classifier or expert/anti-expert ensemble that produces the bias remains opaque, and a small per-token bias may be the aggregate of many contributions inside that ensemble.

4.3 Distribution-Level Methods

These methods inject external information by interpolating the model's output distribution with an external distribution *after* the forward pass completes. They specialize the intervention operator (Eq. 1) with $z_t = p_\theta(y_t)$, where the additive perturbation takes the form $\delta = \lambda(q - p_\theta)$:

$$p^*(y_t) = (1 - \lambda)\, p_\theta(y_t) + \lambda\, q(y_t), \qquad (5)$$

where q comes from a retrieval datastore or parametric memory and λ is a tunable interpolation weight. Crucially, the base model's forward pass is unmodified; the intervention operates on the output distribution. This distinguishes distribution-level methods from approaches like RAG that modify the model's *input* context (discussed as boundary cases in §4.4).

kNN-LM (Khandelwal et al., 2020) builds a datastore of (key, value) pairs by caching the hidden states and corresponding next tokens from a corpus. At inference time, the model's hidden state $h_{L,t}$ is used to retrieve the k nearest neighbors from this datastore, producing a distribution $q(y_t)$ via softmax over negative distances. The final distribution is the interpolation in Eq. 5. Because the datastore can be swapped without retraining, kNN-LM enables domain adaptation by simply changing which corpus is indexed, achieving a 2.9-point perplexity improvement on Wikitext-103 with no additional training.

The Continuous Cache (Grave et al., 2017) stores recent hidden activations and interpolates a cache distribution with the model's output via a learned gate. Memory Decoder (Cao et al., 2025) and MLP Memory (Wei et al., 2026) replace explicit retrieval with parametric modules, achieving faster inference while maintaining the distribution-interpolation interface.

Stability. Because p^* is a convex combination of two valid distributions, it is itself a valid distribution for any $\lambda \in [0, 1]$ and any q, and $\mathrm{TV}(p^*, p_\theta) \leq \lambda$ under the convention $\mathrm{TV}(p, q) = \frac{1}{2}\|p - q\|_1$. Perturbation magnitude is therefore controlled by λ, while behavioral reliability depends on the quality and calibration of q: a mismatched datastore injects systematic error even at small λ, and a fixed λ cannot adapt to per-step retrieval confidence. Moderate values (0.1–0.3) typically ground knowledge without destabilizing fluency (Khandelwal et al., 2020); Continuous Cache, Memory Decoder, and MLP Memory address the fixed-λ limit by replacing the static weight with an input-dependent gate.

Interpretability. Distribution interpolation supports a form of interpretability unavailable to latent or logit methods: provenance through retrieved exemplars. For kNN-LM, each prediction can be traced to the training datapoints whose hidden states were nearest, exposing the source text rather than a learned dictionary direction or a per-token bias. The interpretation is partial: exemplars explain what q contributes but not how q and p_θ combine to produce the final token, since the base model's contribution remains opaque. Parametric memories (Memory Decoder, MLP Memory) trade exemplar transparency for inference speed.

4.4 Boundary Cases and Baselines

Decoding-level methods modify the selection rule $\mathcal{D}$ rather than the predictive state, so they do not fit Eq. 1 and ρ does not apply: contrastive decoding (Li et al., 2023b) favors tokens where an expert disagrees with an amateur, and NeuroLogic A*esque decoding (Lu et al., 2022) enforces hard lexical constraints via lookahead search. Prompting baselines (chain-of-thought (Wei et al., 2022), tree-of-thought (Yao et al., 2023)) are essential reference points (§5) but do not modify internal states. Other related families fall outside scope by violating tunability or reversibility (prefix/prompt tuning, ReFT, weight editing) or by intervening on the input rather than the predictive state (RAG); see App. F for full criteria.

5 Challenges and Future Directions

5.1 Practitioner Guidance and Failure Modes

Table 12 (Appendix J) summarizes the steering landscape from a practitioner's view: rows are intervention families, columns group deployment (access & aux models, cost), mechanism (reach, granularity, interpretability/provenance), and practitioner judgment (best-suited tasks with their typical failure modes, when to prefer the family over prompting). The remainder of this section motivates the entries.

On the AxBench settings evaluated by Wu et al. (2025b) (Gemma-2-2B/9B, concept-level tasks), prompting outperforms the tested representation-based steering methods, with finetuning next; this ordering is corroborated by human evaluation (Studdiford et al., 2025). SAEBench (Karvonen et al., 2025) adds that common SAE proxy metrics do not reliably predict downstream steering performance.

We therefore frame steering not as a replacement for prompting, but as a complementary control primitive with a different systems profile: direct, tunable, reversible interventions on model computation; support for causal/mechanistic analysis; and access to control regimes that are brittle, entangled, or unreachable through prompts alone. Mishra et al. (2026) formalize this distinction by showing that activation steering can produce residual-stream states unreachable by any discrete prompt.

Four failure modes recur. *Distribution shift*: unconstrained ρ pushes activations off the training manifold, causing collapse. *Control vs. capability trade-off*: even bounded ρ erodes general capabilities as steering strength rises (§3.3). *Hyperparameter sensitivity*: optimal coefficient, layer, and feature choices rarely transfer across tasks or models, forcing per-deployment tuning. *Safety fragility*: refusal is often mediated by a single direction that can be ablated trivially (Arditi et al., 2024), and even benign steering vectors have been shown to increase jailbreak risk as a side effect (Xiong et al., 2026). **Practitioner guidance:** treat prompting as the default; prefer supervised over contrastive signals where labeled data is available (Li et al., 2023a); audit for side effects; and consider unified toolkits such as EasyEdit2 (Xu et al., 2025).

Deployment access. Intervention location also determines access requirements: latent methods generally require white-box access to hidden states and hooks into the forward pass; logit methods require access to output scores and often auxiliary models; distribution-level methods require retrieval or memory infrastructure; and decoding-only methods can sometimes be implemented with black-box scoring or constrained generation APIs. These constraints often dominate method choice in practice.

5.2 Open Problems

The most pressing open problem is the *geometry of safe perturbation*: a characterization of the activation manifold that would predict whether a given ρ preserves capabilities or causes collapse. Initial steps exist (Taimeskhanov et al., 2026), but a comprehensive theory remains open: at matched ρ, the direction of δ also matters. Two further problems compound this gap: *composability*, since practical deployments need safety, style, and factuality controls simultaneously yet few works rigorously evaluate composition across heterogeneous steering families and deployment settings; and *safety governance*, namely whether safety can be made geometrically complex enough to resist steering-based attacks without sacrificing beneficial steerability, likely by distributing safety across many entangled features rather than single removable directions.

Adjacent directions. Three threads outside this survey's scope are natural extensions of the framework. Multimodal steering (Wu et al., 2025a) applies the same operator to vision-language activations but adds modality-specific manifold concerns. Mechanistic-interpretability tools (circuit discovery, feature flow (Laptev et al., 2025)) increasingly serve as δ-construction methods, blurring the line between interpretation and control. And weight editing (ROME, MEMIT), excluded here for modifying base weights, shares the predictive-state view and may admit a unified treatment under a generalized δ on parameters rather than activations. Bridging these threads with the location-conditioned framework is a productive direction for future surveys.

6 Conclusion

This survey unifies inference-time steering under an intervention operator $z'_t = z_t + \delta$ applied at one of three *intervention locations* (latent, logit, external-memory distribution interpolation), with decoding-level methods treated as boundary cases. The perturbation ratio ρ provides a location-conditioned diagnostic statistic. Latent and logit methods exhibit complementary strengths, prompting baselines remain competitive on concept-level tasks, and bounding ρ correlates with capability retention. Steering is best framed as a complementary control primitive with a different systems profile: direct, tunable, reversible interventions that are categorically distinct from prompting (Mishra et al., 2026). The field's most pressing needs are a geometric theory of safe perturbation, composability studies, and governance frameworks for the dual-use nature of representation control.

Limitations

This survey has several limitations. Our coverage is restricted to text-generation steering; multimodal steering (Wu et al., 2025a) and non-generative applications are out of scope. The unified framework centers on intervention location and ρ. Because ρ is defined by location-specific quantities (Euclidean ratio for vector-valued states, total variation for distributions, undefined for decoding), it is not commensurate across families and we therefore do not use it for cross-family ranking. Secondary axes such as computational cost, data requirements, and deployment constraints receive less systematic treatment. The case study in Appendix H is illustrative, not a benchmark: it uses a single model family (Qwen2.5), keyword-based judges, and 200 samples per task; absolute scores depend on those choices and should not be read as head-to-head benchmark results. The field is evolving rapidly: papers published after our literature cutoff (May 2026) are not covered, and the benchmark landscape (AxBench, SAEBench) is itself still maturing.

Use of AI assistants. AI writing assistants were used for language editing and literature searching during the preparation of this manuscript. All generated content was verified, revised, and approved by the authors, who take full responsibility for the final text.

References

Hyeseon An, Shinwoo Park, Hyundong Jin, and Yo-Sub Han. 2026. Steering language models before they speak: Logit-level interventions. *Preprint*, arXiv:2601.10960.

Sandro Andric. 2025. Brain-grounded axes for reading and steering llm states. *Preprint*, arXiv:2512.19399.

Anthropic. 2023. Towards monosemanticity: Decomposing language models with dictionary learning. `https://transformer-circuits.pub/2023/monosemantic-features/index.html`. Accessed 2026-03-11.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, and 32 others. 2022. Constitutional ai: Harmlessness from ai feedback. *Preprint*, arXiv:2212.08073.

Lukasz Bartoszcze, Sarthak Munshi, Bryan Sukidi, Jennifer Yen, Zejia Yang, David Williams-King, Linh Le, Kosi Asuzu, and Carsten Maple. 2025. Representation engineering for large language models: Survey and research challenges. *Preprint*, arXiv:2502.17601.

Daniel Beaglehole, Adityanarayanan Radhakrishnan, Enric Boix-Adserà, and Mikhail Belkin. 2026. Toward universal steering and monitoring of ai models. *Science*, 391(6787):787–792.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, and 12 others. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*.

Madeline Brumley, Joe Kwon, David Krueger, Dmitrii Krasheninnikov, and Usman Anwar. 2024. Comparing bottom-up and top-down steering approaches on in-context learning tasks. In *NeurIPS 2024 Workshop on Foundation Model Interventions (MINT)*.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2023. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations*.

Jiaqi Cao, Jiarui Wang, Rubin Wei, Qipeng Guo, Kai Chen, Bowen Zhou, and Zhouhan Lin. 2025. Memory decoder: A pretrained, plug-and-play memory for large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Sviatoslav Chalnev, Matthew Siu, and Arthur Conmy. 2024. Improving steering vectors by targeting sparse autoencoder features. *Preprint*, arXiv:2411.02193.

Seonglae Cho, Zekun Wu, and Adriano Koshiyama. 2026. Corrsteer: Generation-time llm steering via correlated sparse autoencoder features. In *International Conference on Machine Learning (ICML)*.

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, Parker Schuh, Kensen Shi, Sasha Tsvyashchenko, Joshua Maynez, Abhishek Rao, Parker Barnes, Yi Tay, Noam Shazeer, and 1 others. 2023. Palm: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24(240):1–113.

Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2022. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8493–8502. Association for Computational Linguistics.

Quy-Anh Dang and Chris Ngo. 2026. Selective steering: Norm-preserving control through discriminative layer selection. *Preprint*, arXiv:2601.19375.

Sumanth Dathathri, Andrea Madotto, Janice Lan, Jane Hung, Eric Frank, Piero Molino, Jason Yosinski, and Rosanne Liu. 2020. Plug and play language models: A simple approach to controlled text generation. In *International Conference on Learning Representations*.

Xiangjue Dong, Maria Teleki, and James Caverlee. 2024. A survey on llm inference-time self-improvement. *Preprint*, arXiv:2412.14352.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Shauna Kravec, Tom Conerly, Nova DasSarma, Sheer El Showk, Eli Tran-Johnson, Stanislav Fort, Zac Hatfield-Dodds, Danny Hernandez, Dawn Drain, Liane Lovitt, Kamal Ndousse, and 4 others. 2021. A mathematical framework for transformer circuits. `https://transformer-circuits.pub/2021/framework/index.html`. Accessed 2026-03-11.

Joshua Engels, Eric J. Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. 2025. Not all language model features are one-dimensionally linear. In *International Conference on Learning Representations (ICLR)*.

Soham Gadgil, Chris Lin, and Su-In Lee. 2026. Where to steer: Input-dependent layer selection for steering improves llm alignment. *Preprint*, arXiv:2604.03867.

Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2025. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations (ICLR)*.

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2021. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.

Navita Goyal and Hal Daumé III. 2026. Steering safely or off a cliff? rethinking specificity and robustness in inference-time interventions. In *Proceedings of the 2026 Conference of the European Chapter of the Association for Computational Linguistics*. Association for Computational Linguistics.

Edouard Grave, Armand Joulin, and Nicolas Usunier. 2017. Improving neural language models with a continuous cache. In *International Conference on Learning Representations (ICLR)*.

Pengrui Han, Xueqiang Xu, Keyang Xuan, Peiyang Song, Siru Ouyang, Runchu Tian, Yuqing Jiang, Cheng Qian, Pengcheng Jiang, Jiashuo Sun, Junxia Cui, Ming Zhong, Ge Liu, Jiawei Han, and Jiaxuan You. 2026. Steer2adapt: Dynamically composing steering vectors elicits efficient adaptation of llms. In *ICLR 2026 Workshop Re-Align (Long Paper)*.

Ruben Härle, Felix Friedrich, Manuel Brack, Björn Deiseroth, Patrick Schramowski, and Kristian Kersting. 2024. Scar: Sparse conditioned autoencoders for concept detection and steering in llms. In *NeurIPS 2024 Workshop on Socially Responsible Language Modelling Research (SoLaR)*.

Zirui He, Haiyan Zhao, Yiran Qiao, Fan Yang, Ali Payani, Jing Ma, and Mengnan Du. 2025. Saif: A sparse autoencoder framework for interpreting and steering instruction following of language models. *Preprint*, arXiv:2502.11356.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Damjan Kalajdzievski. 2025. The logical implication steering method for conditional interventions on transformer generation. In *Proceedings of the Forty-second International Conference on Machine Learning*.

Adam Karvonen, Can Rager, Johnny Lin, Curt Tigges, Joseph Isaac Bloom, David Chanin, Yeu-Tong Lau, Eoin Farrell, Callum Stuart McDougall, Kola Ayonrinde, Demian Till, Matthew Wearden, Arthur Conmy, Samuel Marks, and Neel Nanda. 2025. Saebench: A comprehensive benchmark for sparse autoencoders in language model interpretability. In *Proceedings of the Forty-second International Conference on Machine Learning*.

Nitish Shirish Keskar, Bryan McCann, Lav R. Varshney, Caiming Xiong, and Richard Socher. 2019. Ctrl: A conditional transformer language model for controllable generation. *Preprint*, arXiv:1909.05858.

Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. 2020. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations*.

Kai Konen, Sophie Jentzsch, Diaoulé Diallo, Peer Schütt, Oliver Bensch, Roxanne El Baff, Dominik Opitz, and Tobias Hecking. 2024. Style vectors for steering generative large language models. In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 782–802. Association for Computational Linguistics.

Ben Krause, Akhilesh Deepak Gotmare, Bryan McCann, Nitish Shirish Keskar, Shafiq Joty, Richard Socher, and Nazneen Fatema Rajani. 2021. GeDi: Generative discriminator guided sequence generation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 4929–4952. Association for Computational Linguistics.

Lucio La Cava and Andrea Tagarelli. 2025. Toward preference-aligned large language models via residual-based model steering. *Preprint*, arXiv:2509.23982.

Daniil Laptev, Nikita Balagansky, Yaroslav Aksenov, and Daniil Gavrilov. 2025. Analyze feature flow to enhance interpretation and steering in language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3045–3059. Association for Computational Linguistics.

Kenneth Li, Oam Patel, Fernanda Viegas, Hanspeter Pfister, and Martin Wattenberg. 2023a. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*.

Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. 2023b. Contrastive decoding: Open-ended text generation as optimization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12286–12312. Association for Computational Linguistics.

Xiang Lisa Li and Percy Liang. 2021. Prefix-tuning: Optimizing continuous prompts for generation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4582–4597. Association for Computational Linguistics.

Alisa Liu, Maarten Sap, Ximing Lu, Swabha Swayamdipta, Chandra Bhagavatula, Noah A. Smith, and Yejin Choi. 2021. Dexperts: Decoding-time controlled text generation with experts and anti-experts. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 6691–6706. Association for Computational Linguistics.

Ximing Lu, Sean Welleck, Peter West, Liwei Jiang, Jungo Kasai, Daniel Khashabi, Ronan Le Bras, Lianhui Qin, Youngjae Yu, Rowan Zellers, Noah A. Smith, and Yejin Choi. 2022. Neurologic a*esque decoding: Constrained text generation with lookahead heuristics. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 780–799. Association for Computational Linguistics.

Jinqi Luo, Tianjiao Ding, Kwan Ho Ryan Chan, Darshan Thaker, Aditya Chattopadhyay, Chris Callison-Burch, and Rene Vidal. 2024. Pace: Parsimonious concept engineering for large language models. In *Advances in Neural Information Processing Systems*.

Samuel Marks and Max Tegmark. 2023. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in gpt. In *Advances in Neural Information Processing Systems*.

Aayush Mishra, Daniel Khashabi, and Anqi Liu. 2026. Steered llm activations are non-surjective. *Preprint*, arXiv:2604.09839. ICLR 2026 Workshop.

Lucas Monteiro Paes, Nivedha Sivakumar, Yinong Oliver Wang, Masha Fedzechkina, Barry-John Theobald, Luca Zappella, and Nicholas Apostoloff. 2026. Dso: Direct steering optimization for bias mitigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Aashiq Muhamed, Jacopo Bonato, Mona T. Diab, and Virginia Smith. 2025. Saes can improve unlearning: Dynamic sparse autoencoder guardrails for precision unlearning in llms. In *ICML 2025 Workshop on Reliable and Responsible Foundation Models*.

Matthew Lyle Olson, Neale Ratzlaff, Musashi Hinck, Shao-yen Tseng, and Vasudev Lal. 2024. Steering large language models to evaluate and amplify creativity. In *NeurIPS 2024 Workshop on Creativity & Generative AI (Spotlight)*.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*.

Kiho Park, Yo Joong Choe, and Victor Veitch. 2023. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*.

Sungjoon Park, Varun Ramamurthi, and Owen Terry. 2025. Neologism learning as a parameter-efficient alternative to fine-tuning for model steering. *Preprint*, arXiv:2512.18551.

Shrimai Prabhumoye, Alan W. Black, and Ruslan Salakhutdinov. 2020. Exploring controllable text generation techniques. In *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*.

Yifu Qiu, Zheng Zhao, Yftah Ziser, Anna Korhonen, Edoardo M. Ponti, and Shay B. Cohen. 2024. Spectral editing of activations for large language model alignment. In *Advances in Neural Information Processing Systems*.

Gorjan Radevski, Kiril Gashteovski, Giwon Hong, Carolin Lawrence, and Goran Glavaš. 2026. Compositional steering of large language models with steering tokens. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. 2024. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522. Association for Computational Linguistics.

Daniel Scalena, Gabriele Sarti, and Malvina Nissim. 2024. Multi-property steering of large language models with dynamic activation composition. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 577–603. Association for Computational Linguistics.

Samuel Soo, Chen Guang, Wesley Teng, Chandrasekaran Balaganesh, Tan Guoxian, and Yan Ming. 2025. Interpretable steering of large language models with feature guided activation additions. In *ICLR 2025 Workshop on Building Trust in Language Models and Applications (BuildingTrust)*.

Asa Cooper Stickland, Alexander Lyzhov, Jacob Pfau, Salsabila Mahdi, and Samuel R. Bowman. 2024. Steering without side effects: Improving post-deployment control of language models. In *NeurIPS 2024 Workshop on Safe Generative AI (SafeGenAI)*.

Alessandro Stolfo, Vidhisha Balachandran, Safoora Yousefi, Eric Horvitz, and Besmira Nushi. 2025. Improving instruction-following in language models through activation steering. In *International Conference on Learning Representations*.

Zach Studdiford, Timothy T. Rogers, Siddharth Suresh, and Kushin Mukherjee. 2025. Evaluating steering techniques using human similarity judgments. *Preprint*, arXiv:2505.19333.

Jiuding Sun, Sidharth Baskaran, Zhengxuan Wu, Michael Sklar, Christopher Potts, and Atticus Geiger. 2025. Hypersteer: Activation steering at scale with hypernetworks. *Preprint*, arXiv:2506.03292.

Magamed Taimeskhanov, Samuel Vaiter, and Damien Garreau. 2026. Towards understanding steering strength. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. Llama: Open and efficient foundation language models. *Preprint*, arXiv:2302.13971.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2024. Steering language models with activation engineering. *Preprint*, arXiv:2308.10248.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30.

Hieu M. Vu and Tan M. Nguyen. 2025. Angular steering: Behavior control via rotation in activation space. In *Advances in Neural Information Processing Systems*. Spotlight.

Anyi Wang, Xuansheng Wu, Dong Shu, Yunpu Ma, and Ninghao Liu. 2025a. Enhancing llm steering through sparse autoencoder-based vector refinement. *Preprint*, arXiv:2509.23799.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. 2023. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*.

Mengru Wang, Xingyu Chen, Yue Wang, Zhiwei He, Jiahao Xu, Tian Liang, Qiuzhi Liu, Yunzhi Yao, Wenxuan Wang, Ruotian Ma, Haitao Mi, Ningyu Zhang, Zhaopeng Tu, Xiaolong Li, and Dong Yu. 2025b. Two experts are all you need for steering thinking: Reinforcing cognitive effort in moe reasoning models without additional training. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Mengru Wang, Ziwen Xu, Shengyu Mao, Shumin Deng, Zhaopeng Tu, Huajun Chen, and Ningyu Zhang. 2025c. Beyond prompt engineering: Robust behavior control in llms via steering target atoms. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23381–23399. Association for Computational Linguistics.

Weixuan Wang, Jingyuan Yang, and Wei Peng. 2025d. Semantics-adaptive activation intervention for llms via dynamic steering vectors. In *International Conference on Learning Representations (ICLR)*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*.

Rubin Wei, Jiaqi Cao, Jiarui Wang, Jushi Kai, Qipeng Guo, Bowen Zhou, and Zhouhan Lin. 2026. Mlp memory: A retriever-pretrained memory for large language models. In *International Conference on Learning Representations (ICLR)*.

Lyucheng Wu, Mengru Wang, Ziwen Xu, Tri Cao, Nay Oo, Bryan Hooi, and Shumin Deng. 2025a. Automating steering for safe multimodal large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 792–814. Association for Computational Linguistics.

Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. 2025b. Axbench: Steering llms? even simple baselines outperform sparse autoencoders. In *Proceedings of the Forty-second International Conference on Machine Learning*.

Zhengxuan Wu, Aryaman Arora, Zheng Wang, Atticus Geiger, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. 2024. Reft: Representation finetuning for language models. In *Advances in Neural Information Processing Systems*.

Zhengxuan Wu, Qinan Yu, Aryaman Arora, Christopher D. Manning, and Chris Potts. 2025c. Improved representation steering for language models. In *Advances in Neural Information Processing Systems (NeurIPS, Spotlight)*.

Jiaqing Xie. 2025. A comparative analysis of sparse autoencoder and activation difference in language model steering. *Preprint*, arXiv:2510.01246.

Chen Xiong, Zhiyuan He, Pin-Yu Chen, Ching-Yun Ko, and Tsung-Yi Ho. 2026. Steering externalities: Benign activation steering unintentionally increases jailbreak risk for large language models. *Preprint*, arXiv:2602.04896.

Ziwen Xu, Shuxun Wang, Kewei Xu, Haoming Xu, Mengru Wang, Xinle Deng, Yunzhi Yao, Guozhou Zheng, Huajun Chen, and Ningyu Zhang. 2025. Easyedit2: An easy-to-use steering framework for editing large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 522–535. Association for Computational Linguistics.

Jingyuan Yang, Rongjun Li, Weixuan Wang, Ziyu Zhou, Zhiyong Feng, and Wei Peng. 2025. Lf-steering: Latent feature activation steering for enhancing semantic consistency in large language models. *Preprint*, arXiv:2501.11036.

Kevin Yang and Dan Klein. 2021. FUDGE: Controlled text generation with future discriminators. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3511–3535. Association for Computational Linguistics.

Shu Yang, Shenzhe Zhu, Liang Liu, Lijie Hu, Mengdi Li, and Di Wang. 2024. Exploring the personality traits of llms through latent features steering. *Preprint*, arXiv:2410.10863.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*.

Li-Ming Zhan, Bo Liu, Zexin Lu, Yujie Feng, Chengqiang Xie, Jiannong Cao, and Xiao-Ming Wu. 2025. Deal: Disentangling transformer head activations for llm steering. In *ACL 2025 Workshop KnowFM (Non-archival)*.

Li-Ming Zhan, Bo Liu, Chengqiang Xie, Jiannong Cao, and Xiao-Ming Wu. 2026. Real: Reading out transformer activations for precise localization in language model steering. In *International Conference on Learning Representations (ICLR)*.

Hanqing Zhang, Haolin Song, Shaoyu Li, Ming Zhou, and Dawei Song. 2023. A survey of controllable text generation using transformer-based pre-trained language models. *ACM Computing Surveys*.

Hengyuan Zhang, Zhihao Zhang, Mingyang Wang, and 1 others. 2026. Locate, steer, and improve: A practical survey of actionable mechanistic interpretability in large language models. *Preprint*, arXiv:2601.14004. ArXiv preprint, 29 authors.

Ningyu Zhang, Yunzhi Yao, Bozhong Tian, Peng Wang, Shumin Deng, Mengru Wang, Zekun Xi, Shengyu Mao, Jintian Zhang, Yuansheng Ni, Siyuan Cheng, Ziwen Xu, Xin Xu, Jia-Chen Gu, Yong Jiang, Pengjun Xie, Fei Huang, Lei Liang, Zhiqiang Zhang, and 3 others. 2024. A comprehensive study of knowledge editing for large language models. *Preprint*, arXiv:2401.01286.

Haiyan Zhao, Xuansheng Wu, Fan Yang, Bo Shen, Ninghao Liu, and Mengnan Du. 2026a. Denoising concept vectors with sparse autoencoders for improved language model steering. In *Findings of the Association for Computational Linguistics: EACL 2026*.

Hongjue Zhao, Haosen Sun, Jiangtao Kong, Xiaochang Li, Qineng Wang, Liwei Jiang, Qi Zhu, Tarek Abdelzaher, Yejin Choi, Manling Li, and Huajie Shao. 2026b. Odesteer: A unified ode-based steering framework for llm alignment. In *International Conference on Learning Representations*.

Yu Zhao, Alessio Devoto, Giwon Hong, Xiaotang Du, Aryo Pradipta Gema, Hongru Wang, Xuanli He, Kam-Fai Wong, and Pasquale Minervini. 2025. Steering knowledge selection behaviours in llms via sae-based representation engineering. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5117–5136. Association for Computational Linguistics.

Rongyi Zhu, Yuhui Wang, Tanqiu Jiang, Jiacheng Liang, and Ting Wang. 2025. Self-improving model steering. *Preprint*, arXiv:2507.08967.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, and 2 others. 2023. Representation engineering: A top-down approach to ai transparency. *Preprint*, arXiv:2310.01405.

A Full Taxonomy Table

Table 2: Comprehensive mapping of representative steering methods to design axes. *Location*: where the intervention is applied. *Signal Source*: how the steering signal is computed. *Granularity*: behavioral level of control. *Interp.*: interpretability of control signals.

Method	**Venue**	**Location**	**Signal Source**	**Granularity**	**Interp.**
Training-free contrastive extraction (§4.1)					
ActAdd (Turner et al., 2024)	arXiv '24	Latent	None	Concept	Medium
CAA (Rimsky et al., 2024)	ACL '24	Latent	None	Concept	Medium
RepE (Zou et al., 2023)	arXiv '23	Latent	None	Concept	Medium
Training-free contrastive extraction: style & instruction (§4.1)					
Style Vectors (Konen et al., 2024)	EACL-F '24	Latent	None	Style	Medium
Instr. Steering (Stolfo et al., 2025)	ICLR '25	Latent	None	Instruction	Medium
Bottom-up/Top-down (Brumley et al., 2024)	NeurIPS-W '24	Latent	None	Concept	Medium
Supervised probe directions (§4.1)					
ITI (Li et al., 2023a)	NeurIPS '23	Latent	Probe	Feature	High
SEA (Qiu et al., 2024)	NeurIPS '24	Latent	Rep-learn	Concept	Med–High
Learned sparse features (§4.1)					
SAE-TS (Chalnev et al., 2024)	arXiv '24	Latent	SAE+opt	Feature	High
FGAA (Soo et al., 2025)	ICLR-W '25	Latent	SAE+opt	Feature	High
CorrSteer (Cho et al., 2026)	ICML '26	Latent	SAE+corr	Adaptive	High–Med
LF-Steering (Yang et al., 2025)	arXiv '25	Latent	SAE+task	Feature	High
PaCE (Luo et al., 2024)	NeurIPS '24	Latent	SAE+proj	Feature	High
SpARE (Zhao et al., 2025)	NAACL '25	Latent	SAE+MI	Feature	High
DSG (Muhamed et al., 2025)	ICML-W '25	Latent	SAE+clf	Feature	High
SDCV (Zhao et al., 2026a)	EACL-F '26	Latent	SAE+denoise	Feature	High
SAE-RSV (Wang et al., 2025a)	arXiv '25	Latent	SAE+refine	Feature	High
SAIF (He et al., 2025)	arXiv '25	Latent	SAE+interp	Feature	High
SCAR (Härle et al., 2024)	NeurIPS-W '24	Latent	SAE+cond	Feature	High
STA (Wang et al., 2025c)	ACL '25	Latent	SAE+atoms	Feature	High
Personality (Yang et al., 2024)	arXiv '24	Latent	SAE+task	Feature	High
Auxiliary classifiers and expert ensembles (§4.2)					
GeDi (Krause et al., 2021)	EMNLP-F '21	Logit	Aux-clf	Token	Med–High
FUDGE (Yang and Klein, 2021)	NAACL '21	Logit	Aux-clf	Token	Medium
DExperts (Liu et al., 2021)	ACL '21	Logit	Aux-clf	Token	Medium
Logit-level (An et al., 2026)	arXiv '26	Logit	Stats	Token	Medium
Stability-oriented methods (§4.1)					
Angular Steer. (Vu and Nguyen, 2025)	NeurIPS '25	Latent	Calibration	Concept	Medium
Selective Steer. (Dang and Ngo, 2026)	arXiv '26	Latent	Disc. select	Adaptive	Medium
ODESteer (Zhao et al., 2026b)	ICLR '26	Latent	Act. stats	Adaptive	Medium
KL-then-Steer (Stickland et al., 2024)	NeurIPS-W '24	Latent	KL-reg	Concept	Medium
Multi-property and adaptive steering (§4.1)					
Dyn. Act. Comp. (Scalena et al., 2024)	BlackboxNLP '24	Latent	Info-theory	Multi	Medium
Comp. Tokens (Radevski et al., 2026)	ACL '26	Latent	Learned	Multi	Medium
Steer2Adapt (Han et al., 2026)	ICLR-W '26	Latent	Learned	Adaptive	Medium
SADI (Wang et al., 2025d)	ICLR '25	Latent	Semantics	Adaptive	Medium
Where-to-Steer (Gadgil et al., 2026)	arXiv '26	Latent	Input-dep	Adaptive	Medium
HyperSteer (Sun et al., 2025)	arXiv '25	Latent	Hypernetwork	Adaptive	Medium
RICE (Wang et al., 2025b)	NeurIPS '25	Latent	MoE-experts	Adaptive	Medium
Human preference alignment (§4.1)					
PaLRS (La Cava and Tagarelli, 2025)	arXiv '25	Latent	Residual	Concept	Medium
SIMS (Zhu et al., 2025)	arXiv '25	Latent	Self-improve	Concept	Medium
RePS (Wu et al., 2025c)	NeurIPS '25	Latent	Bi-direct	Concept	Medium
Safety and robustness (§4.1, §5)					
Steer Externalities (Xiong et al., 2026)	arXiv '26	Latent	Analysis	Concept	Medium
Steer Robustness (Goyal and Daumé III, 2026)	EACL '26	Latent	Analysis	Concept	Medium
Universal Steer. (Beaglehole et al., 2026)	Science '26	Latent	Linear-rep	Concept	High
DEAL (Zhan et al., 2025)	ACL-W '25	Latent	Causal-attr	Feature	High
REAL (Zhan et al., 2026)	ICLR '26	Latent	VQ-AE	Feature	High
DSO (Monteiro Paes et al., 2026)	CVPR '26	Latent	RL+subspace	Concept	Medium
Representation engineering and interpretability (§4.1)					
Feature Flow (Laptev et al., 2025)	ICML '25	Latent	SAE+flow	Feature	High
Brain-Grounded (Andric, 2025)	arXiv '25	Latent	Brain-axes	Feature	High
Creativity (Olson et al., 2024)	NeurIPS-W '24	Latent	Contrastive	Concept	Medium
LIMS (Kalajdzievski, 2025)	ICML '25	Latent	Logic-rules	Concept	High
Steer Strength (Taimeskhanov et al., 2026)	ICML '26	Latent	Theory	Concept	High
Neologism (Park et al., 2025)	arXiv '25	Input	Learned	Concept	Medium
Distribution and memory-based (§4.3)					
kNN-LM (Khandelwal et al., 2020)	ICLR '20	Distribution	Retriever	Token	High
Cont. Cache (Grave et al., 2017)	ICLR '17	Distribution	Cache	Token	High
Mem. Decoder (Cao et al., 2025)	NeurIPS '25	Distribution	Learned	Adaptive	Low–Med
MLP Memory (Wei et al., 2026)	ICLR '26	Distribution	Learned	Adaptive	Low–Med
Decoding-time methods (§4.4)					
Contrastive Dec. (Li et al., 2023b)	ACL '23	Decoding	None	Token/Seq	Low–Med
NeuroLogic A* (Lu et al., 2022)	NAACL '22	Decoding	None	Token/Seq	Medium
Boundary cases (included for comparison; see Appendix B)					
CTRL (Keskar et al., 2019)	arXiv '19	Input	Pretrain	Concept	Medium
PPLM (Dathathri et al., 2020)	ICLR '20	Hybrid	Aux-clf	Concept	Medium
Prefix tuning (Li and Liang, 2021)	ACL '21	Latent	Learned	Concept	Low
Prompt tuning (Lester et al., 2021)	EMNLP '21	Input	Learned	Concept	Low
ReFT (Wu et al., 2024)	NeurIPS '24	Latent	Rep-finetune	Concept	Medium
CoT prompting (Wei et al., 2022)	NeurIPS '22	Input	None	Sequence	Medium
ToT (Yao et al., 2023)	NeurIPS '23	Input/Search	None	Sequence	Medium
Knowl. Neurons (Dai et al., 2022)	ACL '22	Latent	Analysis	Fact	High

B Steering vs. Parameter Updates

	Fine-tune / RLHF	**PEFT (LoRA, prefix)**	**Steering**
Base weights	Updated	Frozen	Frozen
Intervention	Weight update	Learned params	External signal
Tunable at serving	No	Adapter swap/scale	**Yes (α, per request)**
Composable	No	Limited	**Additive**
Reversible	Checkpoint	Swap adapter	**Instant**
Multi-tenant	N copies	N adapters	N **vectors**
Auditable	No	Limited	Varies by location
Min. access	White-box	White-box	**Logit (some)**
Offline cost	High	Moderate	None to moderate
Precision	High	High	Varies

Table 3: Steering vs. parameter-update approaches. The two defining criteria of inference-time steering (§1) are *tunability at serving time via an external per-request signal* and *reversibility per request*; composability follows. Auditability varies by intervention location (§4). Control precision is the main limitation.

C Publication Trends

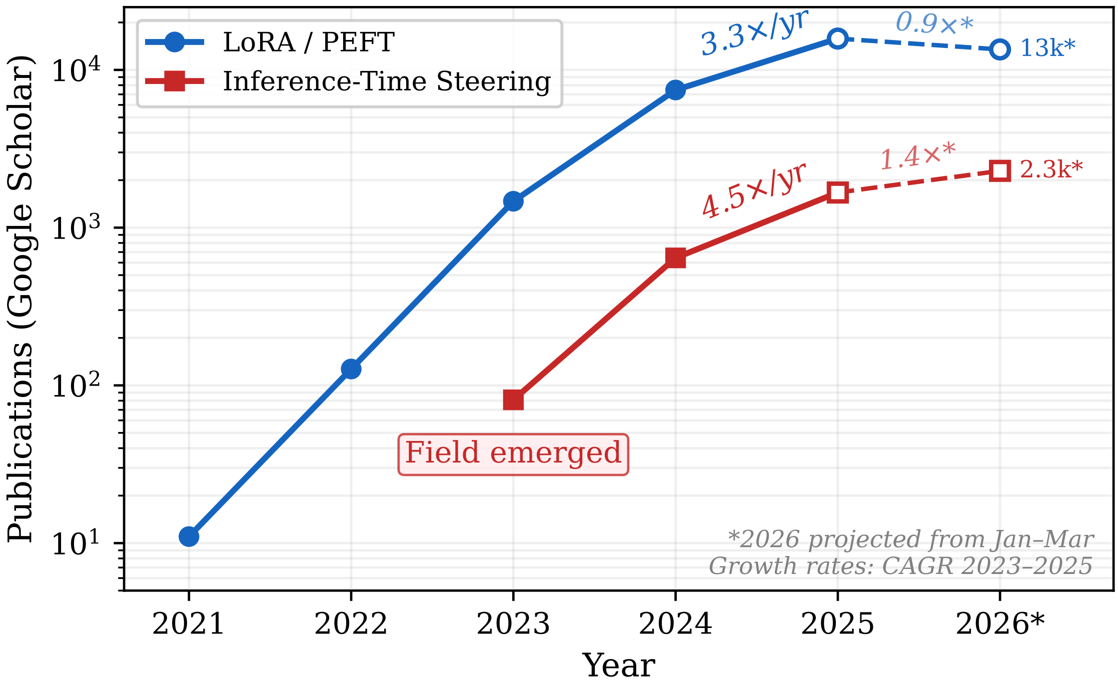

Figure 2: Publication trends for LoRA/PEFT vs. inference-time steering on Google Scholar (queried March 2026, deduplicated by title; indicative only). Dashed lines extrapolate Jan–Mar 2026 counts to the full year.

D Related Surveys

Survey	Focus	What this paper adds
Zhang et al. (2023) (CTG)	Broad controllable generation; training-time + inference-time	Narrower; mechanistic; depth on inference-only post-2023 methods
Zhang et al. (2024) (Knowledge Editing)	Persistent factual update / weight editing	Covers reversible, per-request interventions only; weight editing excluded
Bartoszcze et al. (2025) (RepE)	Latent representation engineering	Cross-location comparison (latent + logit + distribution); access/stability/deployment trade-offs
Zhang et al. (2026) (Actionable MI)	Locate→Steer→Improve; organized around *Interpretable Objects*	Organized around *intervention location*; unifies logit and external-memory distribution interventions with latent steering under one operator
Dong et al. (2024) (Self-Improvement)	Broader inference-time improvement (search, verification, refinement)	Narrower; focuses on direct interventions on the next-token computation of a fixed base LM

Table 4: Positioning against adjacent surveys. Our differentiation is the location-conditioned framework, the ρ stability quantity, and the explicit framing of prompting as the practical baseline rather than as part of the steering family.

E Notation

Symbol	Meaning
θ	Base LM parameters (unmodified during steering)
x	Input context (prompt)
$y_{<t}$	Previously generated tokens $(y_1, \ldots, y_{t-1})$
V	Vocabulary (set of all tokens)
L	Number of transformer layers
d	Hidden state dimension
$h_{\ell,t}$	Hidden state at layer ℓ, position t ($\in \mathbb{R}^d$)
ℓ_t	Logit vector at step t ($\in \mathbb{R}^{\lvert V \rvert}$)
W_{out}	Output projection matrix (maps $h_{L,t}$ to ℓ_t)
τ	Temperature parameter
$\mathcal{D}$	Decoding algorithm
z_t	Predictive state (general: hidden, logit, or distribution)
δ	Steering perturbation (intervention operator)
ϕ	Parameters of the steering method (may be empty)
ρ	Perturbation ratio $\|\delta\|/\|z_t\|$
α	Steering strength coefficient
v	Steering vector ($\in \mathbb{R}^d$)
$e(\cdot), d(\cdot)$	SAE encoder and decoder
s_t	Sparse feature vector $e(h_{\ell,t}) \in \mathbb{R}^k$

Table 5: Mathematical notation used throughout this survey.

F Survey Scope

Status	Method Family	Reason
Included		
✓	Activation steering (CAA, ITI, RepE)	Tunable via α; instantly reversible per request
✓	SAE-based steering (SAE-TS, CorrSteer)	Tunable feature coefficients; reversible per request
✓	Logit guidance (DExperts, FUDGE, GeDi)	Tunable bias strength; reversible per request
✓	External-memory distribution interpolation (kNN-LM, Cont. Cache)	Adjustable interpolation weight λ; datastore swappable per request
Excluded		
×	Full fine-tuning, RLHF	Modifies base weights
×	LoRA, adapters	Learned params; not tunable at serving time
×	Weight editing (ROME, MEMIT)	Modifies base weights
×	Multimodal steering	Out of scope (text generation only)
×	Nucleus sampling, self-consistency	General decoding heuristics; not interventions on the next-token computation of a fixed base LM
×	RAG (input-context retrieval)	Modifies the input, not the predictive state
Boundary cases (discussed but not central)		
∼	Decoding strategies (contrastive dec., NeuroLogic)	Tunable, reversible, but modify $\mathcal{D}$ rather than z_t; ρ undefined
∼	Prefix/prompt tuning	Injects learned vectors but not scalable by α; requires retraining to change
∼	ReFT	Learns intervention params via gradient descent; not runtime-adjustable
∼	PPLM	Hybrid; uses gradient updates during generation
∼	CoT/ToT prompting	No internal state modification; included as baselines

Table 6: Scope of this survey. Included methods produce interventions that are tunable at serving time without retraining. Excluded methods commit to behaviors at training time or modify base weights. Boundary cases share some properties with steering but violate at least one defining criterion (Appendix B).

Curation Methodology

We curated the literature in three passes.

Pass 1: AI-assisted candidate gathering. We used an AI literature-search assistant (acknowledged in Limitations) to aggregate candidate papers in the model-steering space, querying across arXiv, ACL Anthology, OpenReview, and Google Scholar with literature cutoff May 2026. Pass 1 produced 91 candidates organized into nine thematic categories: human-preference alignment, activation-based steering, controlled text generation, model editing and knowledge manipulation, prompt-based control, representation engineering, specialized domain applications, multimodality, and security/jailbreak.

Pass 2: Domain screening. The 91 candidates were screened against this survey's scope (Table 6). We excluded non-NLP applications (autonomous-vehicle steering control, hurricane forecasting, protein engineering, image enhancement, healthcare-explanation steering for clinical decision support) and applications in modalities outside text generation (multimodal/vision-language steering is acknowledged but treated as out of scope; Wu et al., 2025a). Pass 2 produced a working set of NLP/LLM-relevant steering papers used as the bibliographic core.

Pass 3: Criterion screening and supplementation. The Pass-2 set was screened against the two defining criteria (§1): tunability at serving time via an external per-request signal, and per-request reversibility. Methods violating either criterion were either excluded (full fine-tuning, RLHF, LoRA, weight editing, RAG) or retained as boundary cases (decoding strategies, prefix/prompt tuning, ReFT, PPLM, CoT/ToT prompting). We also supplemented the set with foundational pre-2024 references (e.g., kNN-LM, Continuous Cache, GeDi, FUDGE, DExperts) that the AI-assisted Pass 1 underweighted because of citation-recency bias, and with adjacent surveys identified during related-work writing (Table 4, App. D).

Borderline cases. Three categories required explicit judgment. (i) *Prefix and prompt tuning* were excluded as steering methods (learned parameters, not tunable at serving time) but are kept in Table 2 for comparison since they share the latent-injection geometry. (ii) *ReFT* was excluded as steering (learns intervention parameters via gradient descent) but discussed as a boundary case because it operates on representations during inference. (iii) *Decoding-time methods* (contrastive decoding, NeuroLogic) satisfy tunability and reversibility but modify the decoding rule $\mathcal{D}$ rather than the predictive state z_t; we treat them as boundary cases (§4.4) so the operator and ρ analyses remain well-defined.

Final coverage. Table 2 (App. A) covers approximately 50 representative steering methods across the four locations (latent, logit, distribution, decoding) and the boundary categories used for comparison. The full bibliography contains additional foundational and contextual references cited throughout the main text.

G Expressiveness: Algebraic Remark

The expressiveness asymmetry between latent and logit interventions used in §3.3 is best stated as an algebraic observation, not a theorem about steering.

Let $\ell_t = W_{\text{out}} h_{L,t} + b$ be the final-layer logit map. For any logit-level bias $\delta_\ell \in \text{col}(W_{\text{out}})$, the latent edit $\delta_h = W_{\text{out}}^+ \delta_\ell$ at layer L produces an equivalent logit shift via the Moore–Penrose pseudoinverse, since $W_{\text{out}}(h_{L,t} + \delta_h) + b = \ell_t + W_{\text{out}} W_{\text{out}}^+ \delta_\ell = \ell_t + \delta_\ell$. *Assumptions:* the edit is applied at the final layer (no propagation through nonlinear blocks), the bias lies in the column space of W_{out}, and the edit is applied to the post-LayerNorm representation fed to W_{out}. The converse fails: a latent edit δ_h at layer $\ell < L$ propagates through $L - \ell$ nonlinear blocks, producing context-dependent logit changes that no fixed bias can replicate.

This is an algebraic equivalence, not a claim about which intervention is preferable in practice; expressiveness does not entail better steerability, especially on concept-level tasks (§5).

H Illustrative Case Study

Scope of evidence. This appendix presents an *illustrative case study* on a single model family (Qwen2.5) with keyword-based judges and 200 samples per task. Patterns reported here are not benchmark results and do not generalize across model families without further evaluation. Their purpose is to make the framework's quantities (ρ, intervention location, capability retention) concrete on a common setup.

We conduct a controlled comparison of five steering methods on two models (Qwen2.5-3B and 7B), evaluating truthfulness (TruthfulQA), toxicity avoidance (RealToxicityPrompts), and capability retention (MMLU) under identical conditions. We measure truthfulness using the T$\times$I (Truthful $\times$ Informative) metric, which is the product of the fraction of responses judged truthful and the fraction judged informative; the unsteered baseline scores $\sim$0.77–0.78. We instrument each method with the perturbation ratio ρ from §3.3, enabling within-family measurement; cross-family ρ values are not directly comparable. A random-direction baseline controls for perturbation magnitude independent of direction. Table 7 summarizes results at each method's best operating point.

Method	**TruthfulQA (T$\times$I)**	**Toxicity $\downarrow$**	**MMLU**	ρ	ρ **Type**
Qwen2.5-3B-Instruct					
Unsteered	0.770	0.110	0.500	0	—
CAA	0.785	0.000	0.480	0.58	unconstrained
ITI	0.815	0.092	0.505	0.12	unconstrained
SAE Steering	0.795	0.093	0.500	0.15	bounded
Logit Contrast[‡]	0.785	0.008	0.500	0.09	logit
Prompting	0.865	0.105	0.500	N/A	none
Random Dir.	0.650	0.000	0.500	0.58	unconstrained
Qwen2.5-7B-Instruct					
Unsteered	0.780	0.120	0.605	0	—
CAA	0.520	0.000	0.305[†]	2.74	unconstrained
ITI	0.785	0.090	0.610	0.01	unconstrained
SAE Steering	0.570	0.000	0.605	0.66	bounded
Logit Contrast[‡]	0.780	0.018	0.605	0.36	logit
Prompting	0.875	0.097	0.605	N/A	none
Random Dir.	0.000	0.000	0.605	2.67*	unconstrained

Table 7: Controlled comparison of steering methods. T$\times$I and toxicity each report the best value across all tested strengths; ρ corresponds to the T$\times$I operating point. MMLU is reported at the nearest measured strength. [†]CAA on 7B degrades MMLU even at the lowest tested strength (α=0.5). *Random direction on 7B produces T$\times$I=0 at all strengths; ρ shown at α=0.5. [‡]"Logit Contrast" is a simplified DExperts-style baseline using contrastive logit biases rather than full expert/anti-expert LMs; results indicate the family, not DExperts proper. Evaluation uses keyword-based judges; relative ordering is reliable but absolute scores differ from published benchmarks.

Three illustrative patterns emerge in this case study (literature-backed conclusions are flagged separately in §5):

Pattern 1 (case study): ρ tracks capability retention within the latent family. Methods with low ρ (ITI at $\rho \leq 0.12$, Logit Contrast at $\rho \leq 0.36$) maintain MMLU within 1% of baseline on both models. CAA, with ρ=2.74 on 7B even at its lowest tested strength, suffers 50% MMLU degradation and collapses at higher strengths (Figure 3). The random-direction baseline also collapses at matched ρ (T$\times$I = 0.000 on 7B at ρ=2.67), reinforcing that the direction of δ is a second axis to consider alongside ρ.

Pattern 2 (case study): intervention location shapes task fit. On TruthfulQA, latent methods (ITI: 0.785 T$\times$I on 7B) match the Logit Contrast baseline (0.780) while prompting dominates both (0.875). On toxicity, Logit Contrast achieves 85% reduction with zero MMLU loss, outperforming all latent methods at matched capability retention. Logit-level interventions excel where uniform token-level filtering suffices; the evidence for latent methods' advantage on context-dependent tasks is weaker, as prompting outperforms both.

Pattern 3 (case study): ρ distributions as method fingerprints. Figure 4 reveals that per-sample ρ distributions characterize each method: ITI (μ=0.08) concentrates tightly near zero; Logit Contrast

(μ=0.52) spreads moderately below ρ=1.5; SAE Steering (μ=1.5) sits at the stability boundary; and CAA (μ=15.9) and Random Direction (μ=15.5) produce nearly identical distributions spanning ρ=0–80. Cross-family ρ values are not directly commensurate (§3.3); the comparison here is descriptive, not a stability ranking.

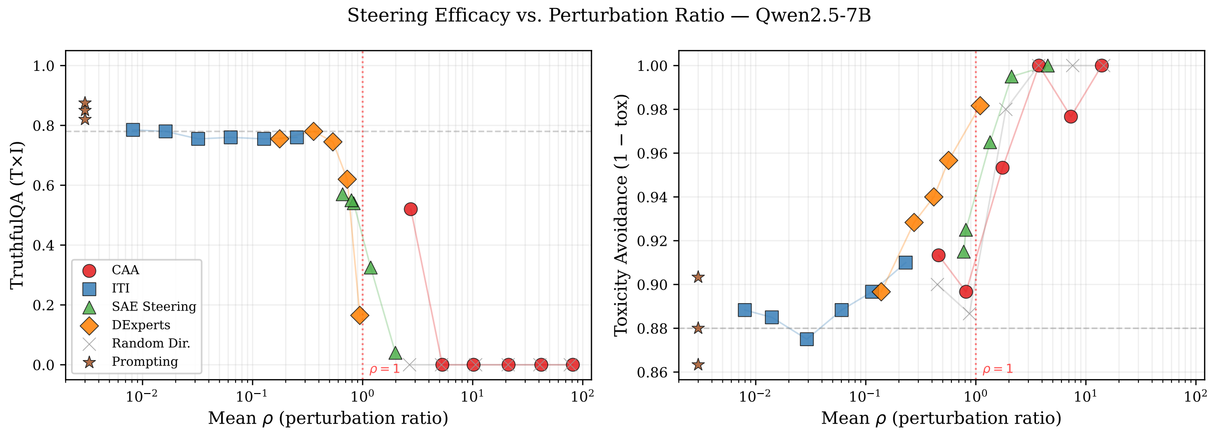

Figure 3: Steering efficacy vs. perturbation ratio ρ on Qwen2.5-7B for TruthfulQA (left) and toxicity avoidance (right). Methods with low ρ (ITI, Logit Contrast) retain high efficacy, while CAA and random directions at $\rho > 1$ collapse to degenerate output. Red dotted line marks $\rho = 1$. Prompting (no ρ) shown at the left edge. Gray dashed line indicates unsteered baseline.

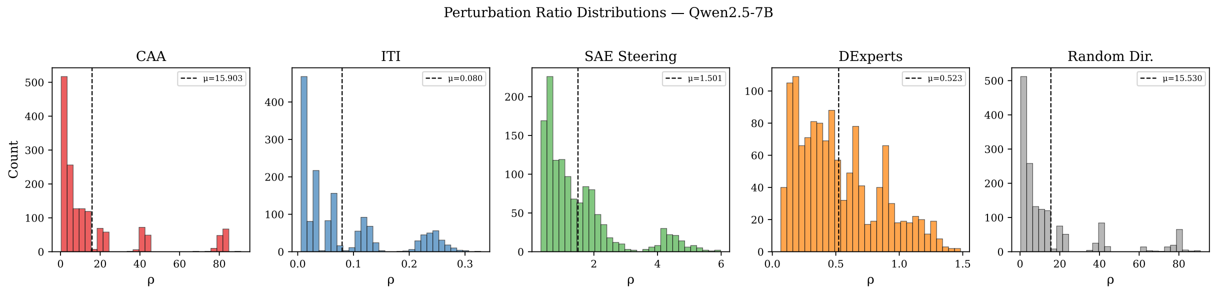

Figure 4: Distribution of perturbation ratio ρ across methods on Qwen2.5-7B, pooled across all tested steering strengths. CAA and random directions produce nearly identical distributions well above $\rho \approx 1$, while ITI and Logit Contrast remain concentrated below it.

Cross-Method Comparison from the Literature

Table 8 aggregates representative reported results from recent papers. These numbers are *not directly comparable* across rows due to differences in base models, evaluation protocols, and metric definitions.

Method	Base Model	TruthfulQA	Toxicity ↓	MMLU (retain)	Access
ITI	Alpaca (LLaMA-7B)	32.5→65.1% T×I	—	Minimal drop	White-box
CAA	Llama-2-13B-chat	Varies by behavior	—	Varies	White-box
ODESteer	LLaMA-3.1-8B	+5.7%	−2.4% toxic	—	White-box
Sel. Steering	Various (9 models)	—	—	0 PPL viol.	White-box
CorrSteer	Gemma-2-2B/LLaMA-3.1-8B	—	+27.2% HarmB.	+3.3% (4k samp.)	White-box
Prompting (AxBench)	Various	*Outperforms SAE/rep-based steering on AxBench tasks*			Any
DExperts	GPT-2-XL	—	Substantial reduction	—	Logit
FUDGE	GPT-2-XL	—	Task-specific	—	Logit
kNN-LM	Transformer-XL	PPL 15.79	—	—	White-box

Table 8: Representative reported results from steering methods (not directly comparable across rows). "T×I" = Truthful × Informative. "HarmB." = HarmBench attack success rate reduction. "Access" = required model access level.

I Method-Family Variants

This appendix groups three within-family comparisons referenced from §4: SAE-based variants, stability-oriented latent methods, and logit-level variants.

I.1 SAE-Based Steering Variants

Method	Feature selection	Key contribution
SAE-TS	Optimize against off-target features	SAE as a lens for measuring side effects
FGAA	Optimize coefficients in SAE space	Steering/coherence trade-off optimization
CorrSteer	Online correlation with task signal	Adaptive; introduces SER metric
LF-Steer	Target inconsistency-related features	Reduces semantic inconsistency in outputs
PaCE	Projection to remove target concept	Disentangled concept removal
SpARE	Mutual information	Identifies features controlling knowledge selection
DSG	Dynamic classifier	Selective suppression of unwanted knowledge
SDCV	Denoising concept vectors	Filters task-irrelevant features
SAE-RSV	Refined steering vectors	Iterative feature refinement

Table 9: SAE-based steering variants. All operate in SAE feature space ($s_t = e(h_{\ell,t})$); they differ in how features are selected and modified.

I.2 Stability-Oriented Methods

Method	ρ mechanism	Key idea
Selective Steer. (Dang and Ngo, 2026)	Norm-preserving rotation	$\|h'\| = \|h\|$ by construction; discriminative layer selection
ODESteer (Zhao et al., 2026b)	Multi-step integration	Small per-step $\|\delta^{(k)}\|$ via step size η; barrier constraints
Angular Steer. (Vu and Nguyen, 2025)	Calibrated rotation	Rotation angle θ tuned to task
KL-then-Steer (Stickland et al., 2024)	KL regularization	Train model to minimize KL between steered/unsteered outputs on benign inputs, then steer

Table 10: Stability-oriented methods. All explicitly bound or reduce ρ-related side effects, addressing the main failure mode of first-generation additive methods (CAA, ActAdd).

I.3 Logit-Level Steering Variants

Method	How δ is computed	Key contribution
DExperts (Liu et al., 2021)	$\alpha(\ell^{\text{exp}} - \ell^{\text{anti}})$	Expert/anti-expert contrast cancels shared linguistic structure
GeDi (Krause et al., 2021)	Bayes-factor ratio from class-conditional LMs	Generative discriminator guidance
FUDGE (Yang and Klein, 2021)	Future-discriminator $P(\text{attr} \mid y_t, y_{<t})$	Predicts future attribute satisfaction
Logit-level (An et al., 2026)	z-normalized log-odds from labeled corpora	Training-free; statistical token score table for multi-task control

Table 11: Logit-level steering variants. All apply a bias δ to ℓ_t; they differ in the source of the bias signal. Unlike latent (white-box) methods, logit-bias interventions can be delivered through APIs that expose per-token bias, but this is a property of logit-level methods specifically rather than a defining criterion of steering.

J Practitioner Decision Table

Table 12: Family-level practitioner decision view (Table 2 gives the per-method view). Cells reflect modal values within each family; individual methods vary. Cost columns use qualitative labels (None / Low / Medium / High). Logit-only access covers APIs that expose per-token biases without activations.

	Deployment		Mechanism			Judgment	
Family	**Access & aux**	**Cost (online / offline / labels)**	**Reach**	**Granul.**	**Interp. / prov.**	**Best suited for**	**When over prompting**
Latent: contrastive / probe (CAA, ActAdd, RepE, ITI)	White-box; no aux (probe for ITI)	Negligible / Low / contrastive pairs	Through $L-\ell$ blocks	Concept	Direction; no per-dim labels	Broad behavioral shifts (refusal, persona, style); collapse at high ρ	Behaviors hard to specify in text; per-request switching
Latent: SAE (SAE-TS, CorrSteer, FGAA, SpARE)	White-box + SAE	Negligible / Med (public SAE)–High / low	Through $L-\ell$ blocks	Feature	SAE feature labels	Fine-grained concept control with provenance; polysemantic features	Need per-feature control or auditability
Logit (DExperts, GeDi, FUDGE)	Logit access; aux classifier or expert+anti-expert LM	+1 forward / Med / per-class labels	Final softmax only	Token	Per-token bias; vocab-inspectable	Token-level filtering (toxicity, format); softmax saturation at high α	Logit-only deployments; uniform token filtering
Distribution (kNN-LM, Cont. Cache, Memory Decoder)	White-box; datastore or parametric memory	$+k$ retrievals / Med–High / corpus	Output mixing post-forward	Token	Retrieved exemplars (highest provenance)	Domain adaptation, factuality; datastore mismatch	Swap corpora without retraining; exemplar-level provenance
Prompting (CoT, instructions, few-shot; baseline)	Black-box, text only; no aux	None / None / none–few examples	N/A (input-side)	Sequence	Natural language	Concept-level tasks (AxBench-style); brittleness, jailbreak surface	Default; reach for steering when prompting is brittle, multi-property, or sub-token

Reading the table. Three patterns guide the practitioner. (i) *Access dictates feasibility before efficacy*: black-box settings collapse the option set to prompting; logit-only adds the logit family; full white-box opens latent, SAE, and distribution-level options. (ii) *Granularity should match task*: token-level filtering favors logit, broad behavioral shifts favor latent, corpus-grounded factuality favors distribution. (iii) *Prompting is the default* (§5); reach for steering when its distinct systems profile pays for the added complexity—per-request switching, multi-tenant control, sub-token or composed control, or refusal-robustness research.